

GazeSpeak: A Low-Cost, LLM-Accelerated Eye Gaze Typing System for ALS Patients

Chaitra B V

Dept. of Computer Science and Design
Dayananda Sagar Academy of Technology and Management
Bengaluru, Karnataka, India
cbelvadi21@gmail.com

Tanvi Kulkarni

Dept. of Computer Science and Design
Dayananda Sagar Academy of Technology and Management
Bengaluru, Karnataka, India
tanvikulkarni1435@gmail.com

Prajwal J

Dept. of Computer Science and Design
Dayananda Sagar Academy of Technology and Management
Bengaluru, Karnataka, India
reachprajwal03@gmail.com

G Manjula

Dept. of Computer Science and Design
Dayananda Sagar Academy of Technology and Management
Bengaluru, Karnataka, India
hod-csd@dsatm.edu.in

Bindhu Madhuri

Dept. of Computer Science and Design
Dayananda Sagar Academy of Technology and Management
Bengaluru, Karnataka, India
bindhumadhuri06@gmail.com

Abstract—Amyotrophic Lateral Sclerosis (ALS) impairs speech and motor control while sparing eye movement, but traditional gaze-based augmentative and alternative communication (AAC) systems are expensive, slow, and reliant on dedicated infrared hardware. This paper presents GazeSpeak, an open-source, webcam-only eye gaze typing system for ALS patients. Operating on consumer hardware without specialized sensors, GazeSpeak integrates MediaPipe-based iris tracking, three-point horizontal calibration, step-based gaze navigation, large language model (LLM)-driven word prediction and abbreviation expansion via Groq-hosted Llama 3.3 70B, a binary-choice conversation engine, rapid-blink emergency SMS alerting, and text-to-speech output. The system provides a modular, multi-modal AAC solution tailored to operate across various stages of ALS disease progression.

Index Terms—Amyotrophic Lateral Sclerosis, Eye Gaze Typing, Augmentative and Alternative Communication, Large Language Models, Assistive Technology, MediaPipe, Webcam-Based Tracking.

I. INTRODUCTION

Amyotrophic Lateral Sclerosis (ALS) is a progressive neurodegenerative disease that destroys motor neurons, gradually stripping patients of the ability to speak, move their limbs, and eventually breathe independently [3], [9]. Because the oculomotor pathways are typically preserved even in advanced disease stages, eye gaze has become the primary input modality for augmentative and alternative communication (AAC) systems used by this population [1], [6].

Despite decades of research, gaze typing remains slow. On-screen keyboards driven by eye trackers typically support fewer than ten words per minute, roughly an order of magnitude below conversational speech [6]. This gap discourages

extended interaction and contributes to the social isolation commonly reported among ALS patients [8]. Two additional obstacles compound the problem. First, many gaze-tracking systems depend on dedicated infrared hardware and lengthy calibration procedures that are expensive, difficult for patients with restricted head control, and prone to drift [2], [5], [14]. Second, research on typing acceleration, low-cost tracking, brain-computer interfaces (BCI), and voice preservation has advanced along mostly separate paths, with comparatively little work integrating these advances into a single deployable tool tailored to ALS [1].

Three specific advances motivate this work. LLM-powered interfaces have demonstrated 29–60% improvements in text-entry rate over baseline keyboards and 57% savings in motor actions [1]. Blink-coded alphabet trees have enabled calibration-free typing on ordinary webcams [2]. Webcam-based deep-learning facial landmark trackers have removed the dependence on infrared hardware entirely [5]. BCI paradigms offer a fallback modality for patients who progress to a fully locked-in state [3], [11].

This paper presents GazeSpeak, an open-source assistive eye gaze typing system that integrates several of these advances into a working, deployable application. The system runs on consumer hardware with a single webcam, uses MediaPipe Face Mesh for iris tracking, provides LLM-accelerated word prediction and abbreviation expansion through Groq-hosted Llama 3.3 70B, supports a binary-choice conversation mode for patients who find letter-by-letter typing too slow, and includes a rapid-blink emergency alert system with SMS notification via Twilio. To address the identified challenges,

our proposed project makes the following five specific contributions:

- 1) A complete, open-source implementation of a webcam-only gaze typing system that requires no infrared hardware or commercial eye tracker.
- 2) A three-point horizontal calibration scheme with IQR-based outlier rejection that adapts gaze zone thresholds to each patient's eye movement range.
- 3) A step-based gaze navigation model with hysteresis-filtered zone transitions, where horizontal eye direction acts as discrete left/right arrow keys rather than continuous cursor control.
- 4) Integration of LLM-based word prediction, context-aware abbreviation expansion, and a binary-choice conversation engine that reduces the motor effort required for communication.
- 5) A rapid-blink emergency alert mechanism using Eye Aspect Ratio (EAR) blink detection, audible alarm, and Twilio SMS notification.

II. RELATED WORK

Cai et al. developed SpeakFaster, an LLM-powered interface for abbreviated text entry that saved 57% more motor actions than traditional predictive keyboards in offline simulation and achieved text-entry rates 29–60% above baseline in lab and field testing with two eye-gaze AAC users with ALS [1]. A companion field study instrumented the everyday gaze-typed communication of an ALS patient over an extended period to characterize real-world conversational patterns [6]. Car'ia surveyed the convergence of BCI spellers with LLMs, tracing the evolution from early statistical language models to contemporary transformer-based predictive writing [7]. Mirbagheri and Chau presented EyeChat, a gaze-controlled web-browsing and chatbot interface for an augmented-reality display, reporting 9.88 words per minute with an LLM-driven response generation system [16].

Lee and Lee proposed a blink-to-alphabet tree that narrows a 26-letter alphabet through three sequential blink-coded decisions and integrates an LLM API for error correction, outperforming three baseline methods in speed and accuracy without calibration [2]. Karatay et al. developed a webcam-only, deep-learning-based facial landmark tracker that eliminates the need for blinking, enabling use by patients who can no longer blink voluntarily [5]. Molina-Cantero et al. reviewed visible-light, low-cost camera eye-tracking methods, cataloguing calibration strategies, head-pose invariance techniques, and the persistent trade-off between affordability and tracking accuracy [14].

Li et al. conducted a PRISMA-guided review of 23 studies applying BCI paradigms to ALS communication, concluding that recent multi-disciplinary efforts have improved decoding precision but that user variability, BCI illiteracy, and fatigue remain open limitations [3]. Peters et al. introduced Inquiry Preview, a multimodal extension of the RSVP Keyboard that fuses switch input with EEG and language-model evidence; a potential end user with incomplete locked-in syndrome typed fastest with the combined modality [11]. An NIH-funded

initiative on a rapid neural typing interface for veterans with ALS reflects continued investment in translating BCI-based typing toward deployment [15].

Ngo et al. released EEGET-ALS, a public EEG and eye-tracking dataset from six ALS patients and 170 healthy controls performing spelling tasks [4]. Liu et al. demonstrated a wireless MXene-based smart contact lens that encodes blink patterns into a resonance-frequency signal, validating biocompatibility in vivo [13]. On the communication support side, Garbey et al. developed AI tools combining pulse and speech analysis to detect emotional responsiveness during ALS clinic visits [8]. Regondi et al. combined voice banking with AI voice-cloning to preserve a patient's vocal identity for AAC devices [9]. Barton et al. surveyed AAC specialists on tele-AAC versus in-person evaluations for ALS patients [10]. A medRxiv study proposed an AI-driven approach to augmentative and assistive communication for ALS patients [12].

III. PROPOSED METHODOLOGY

A. Problem Statement

The gaze tracker runs as a background QThread, capturing webcam frames at 640×480 resolution and processing each frame through MediaPipe's Face Landmarker task. The model detects 478 facial landmarks per frame, including five iris landmarks per eye (indices 468–477). Gaze direction is computed as the horizontal and vertical iris ratio: the position of each iris center relative to the corresponding eye corners, normalized to [0, 1]. The ratios from both eyes are averaged to produce a single gaze coordinate.

A confidence metric is computed from the agreement between the two eyes. When iris landmarks are unavailable (e.g., due to occlusion), a fallback method estimates gaze from eye corner midpoints at reduced confidence. An exponential moving average (EMA) filter smooths the gaze signal before it reaches the navigation controller. The smoothing factor is user-configurable, defaulting to 0.25, which balances responsiveness against jitter.

B. System design

GazeSpeak is a desktop application built in Python, organized into five core architectural modules: vision and tracking, control engine, language engine, outputs and audio, and emergency alerts. Fig. 1 illustrates the end-to-end data flow through the system pipeline. The application utilizes OpenCV for webcam capture, MediaPipe Face Landmarker for 478 3D facial landmarks, PyQt6 for the GUI, Groq-hosted Llama 3.3 70B for LLM capabilities, pyttsx3 for offline TTS synthesis, and AssemblyAI for real-time caregiver speech recognition.

system_design_diagram.jpg

Fig. 1. Data flow through the GazeSpeak AAC pipeline.

C. Three-point horizontal calibration

On each application launch, a full-screen calibration sequence records the patient’s gaze ratio at three horizontal positions: fully left, straight ahead (center), and fully right. At each position, the system collects 60 gaze samples (approximately two seconds at 30 fps), rejects outliers using IQR filtering (samples outside $Q_1 - 1.5 \cdot \text{IQR}$ to $Q_3 + 1.5 \cdot \text{IQR}$ are discarded), and takes the mean of the remaining samples.

The three recorded values (x_{left} , x_{center} , x_{right}) are used to define zone boundaries:

$$\text{screen}_x = \frac{x_{\text{raw}} - x_{\text{left}}}{x_{\text{right}} - x_{\text{left}}} \quad (1)$$

The center point in calibrated space, $c = (x_{\text{center}} - x_{\text{left}}) / (x_{\text{right}} - x_{\text{left}})$, is then used to set asymmetric dead-zone thresholds ($\text{zone}_{\text{left}}$ and $\text{zone}_{\text{right}}$):

$$\text{zone}_{\text{left}} = \max(0, c - \max(0.55 \cdot c, 0.04)) \quad (2)$$

$$\text{zone}_{\text{right}} = \min(1, c + \max(0.45 \cdot (1 - c), 0.12)) \quad (3)$$

The asymmetric margins ensure both zones remain reachable regardless of the patient’s neutral gaze. Calibration is valid only if the left–right spread exceeds 0.05 raw gaze units and the center lies strictly between them; otherwise, recalibration is required.

D. Step-based gaze navigation with hysteresis

Rather than mapping gaze position to a continuous screen cursor, GazeSpeak uses a three-zone, step-based navigation model. Looking left of $\text{zone}_{\text{left}}$ triggers a single leftward step of the highlight. Looking right of $\text{zone}_{\text{right}}$ triggers a single rightward step. Returning to the center zone (between the two

thresholds) stops navigation and arms dwell-time selection: if the patient’s gaze remains centered for a configurable dwell period (default 1200 ms), the currently highlighted key is selected.

To prevent false triggers from noisy gaze data, zone transitions require confirmation over multiple consecutive frames (hysteresis). A zone change is only committed after two consecutive frames (at 30 fps, approximately 66 ms) indicate the same new zone. This filters out transient spikes without adding perceptible delay. The navigation highlight wraps circularly through prediction suggestions and keyboard rows. When the highlight moves past the last key in the last keyboard row, it wraps to the first prediction slot, and vice versa.

E. On-screen keyboard and dwell selection

The keyboard is a QWERTY layout augmented with special keys: SPACE, BACKSPACE, SPEAK (triggers TTS), CLEAR, PHRASES (switches to the quick-phrases panel), and SETTINGS. Key selection occurs through dwell: when the patient returns gaze to the center zone after navigating to the desired key, a countdown timer begins. A visual progress ring fills around the highlighted key. If gaze remains centered for the full dwell duration, the key is pressed. Any zone transition cancels the timer.

F. Word prediction

GazeSpeak combines two prediction paths running in parallel. Local prefix matching: a frequency-ranked dictionary (approximately 300 entries weighted toward medical, comfort, and daily-need vocabulary) provides instant completions as the patient types. Results are sorted by frequency and filtered to exclude exact matches with the current prefix. LLM context-aware prediction: in the background, the current sentence and partial word are sent to Groq-hosted Llama 3.3 70B via the Groq API. The model is prompted to return a JSON array of the five most likely word completions, prioritizing ALS patient communication context (medical needs, comfort, social interaction). When the LLM response arrives, it replaces the local predictions in the UI. Temperature is set to 0.3 for focused predictions.

The two-tier approach ensures the patient always sees suggestions instantly (local), while benefiting from contextually smarter predictions when the LLM response returns (typically within 200–500 ms).

G. Context-aware abbreviation expansion

For patients who prefer to type shorthand, the system detects likely abbreviations using a set of heuristics: the input is short (≤ 5 characters), contains few or no vowels relative to consonants, and does not match a list of common English short words. When an abbreviation is detected, the system sends it along with the full conversation history (caretaker questions and prior patient responses) to Llama 3.3 70B, which returns 2–4 plausible expansions ranked by confidence.

For example, if the caretaker asks “Where did the sound come from?” and the patient types “ftk”, the system might

suggest “from the kitchen”, “from the kids”, or “from the terrace”. Each letter in the abbreviation is treated as potentially the first letter of a word. The LLM response includes confidence scores and reasoning for each expansion, and the suggestions are ranked by confidence before display.

H. Binary-choice conversation engine

For situations where letter-by-letter typing is too slow or fatiguing, GazeSpeak provides a conversation mode. When the caretaker types or speaks a question (via the context bar at the bottom of the screen), the system generates exactly two answer options using Llama 3.3 70B. The options are displayed as two large cards: one on the left side and one on the right side of the screen.

The patient selects an option by looking in the corresponding direction (left or right gaze). After each selection, the engine generates a follow-up pair that narrows the patient’s intended meaning. After up to five rounds of binary narrowing, the LLM composes a natural spoken sentence from the patient’s selection trail and speaks it aloud via TTS. For example, if the caretaker asks “How are you feeling?”, the patient might narrow through “Not great” → “I have pain” → “In my head” across three rounds, and the system speaks the composed sentence “I’m not great. I have pain in my head.”

This mode requires only three gaze gestures to compose a full response, compared to dozens of individual key selections for letter-by-letter typing. The patient can also request regeneration of options, undo a previous selection, or switch to keyboard mode at any point.

I. Caregiver mode and speech-to-text input

The caregiver interacts through a context bar at the bottom of the screen. The caregiver can type a question directly via the physical keyboard or use speech-to-text (AssemblyAI real-time streaming API) to dictate their question verbally. The transcribed or typed question is then fed to both the abbreviation expander (as context) and the conversation engine (as the prompt for binary-choice generation). The application also supports full keyboard and mouse input for caregiver-side configuration, settings adjustment, and recalibration.

J. Text-to-speech output

Composed sentences, quick phrases, and conversation-mode responses are spoken aloud using the pyttsx3 offline TTS engine. Speech rate is user-configurable through the settings panel.

IV. EVALUATION METRICS

Table I summarizes the technology stack. The application is open-source and runs on Windows with Python 3.10+.

A. Rapid-blink emergency alert

The gaze tracker continuously computes the Eye Aspect Ratio (EAR) for both eyes using six landmark points per eye:

$$EAR = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2 \cdot \|p_1 - p_4\|} \quad (4)$$

TABLE I
TECHNOLOGY STACK

Layer	Library	Version
GUI framework	PyQt6	≥6.6.0
Webcam capture	OpenCV	≥4.8.0
Face/iris tracking	MediaPipe	≥0.10.9
Numerical computation	NumPy	≥1.24.0
LLM inference	Groq (Llama 3.3 70B)	≥1.0.0
Text-to-speech	pyttsx3	≥2.90
Emergency SMS	Twilio	≥9.0.0
Speech-to-text	AssemblyAI	≥0.30.0

where p_1 represents the outer eye corner, p_2 and p_3 represent the upper eyelid landmark points, p_4 represents the inner eye corner, and p_5 and p_6 represent the lower eyelid landmark points. A blink is registered when the average EAR across both eyes drops below 0.21 for at least two consecutive frames and then rises above the threshold (i.e., the eyes reopen).

The BlinkAlertManager maintains a sliding window of blink timestamps. If five or more blinks occur within a four-second window, the system triggers an emergency alert: (1) a full-screen red overlay covers the application; (2) a continuous siren alarm plays through the system speaker (alternating frequency sweeps between 800–1200 Hz); (3) an SMS is sent to the configured caregiver phone number via the Twilio API. A 60-second cooldown prevents repeat triggers. Blink detection is automatically paused during calibration and resumes with a two-second grace period to avoid false positives from residual calibration blinks.

The codebase is organized into five Python packages: `gaze/` — gaze tracker (background QThread) and calibration screen; `ui/` — nine PyQt6 widget modules (keyboard, gaze cursor, sentence bar, prediction bar, quick phrases, settings panel, context bar, conversation panel, and emergency screen); `prediction/` — word predictor (local + LLM), abbreviation expander, conversation engine, and the frequency-ranked dictionary; `alerts/` — rapid-blink detection and Twilio SMS emergency alert system; `audio/` — AssemblyAI-based real-time speech-to-text for caregiver input.

Threading model: the gaze tracker runs as a dedicated QThread, emitting Qt signals for gaze updates, frame readiness, tracking loss, and blink events. LLM calls (word prediction, abbreviation expansion, conversation options, and response composition) run in daemon threads to avoid blocking the GUI. Results are marshalled back to the main thread via `QTimer.singleShot(0, ...)` or a dedicated `pyqtSignal`.

The system operates without an internet connection for basic gaze typing, dwell selection, local word prediction, TTS, and emergency alarm. Internet connectivity is required only for LLM-based prediction, abbreviation expansion, conversation mode, SMS alerts, and speech-to-text.

V. RESULTS AND DISCUSSION

A. Feature Comparison with Surveyed Systems

Table II presents a comprehensive feature comparison between the proposed methodology and state-of-the-art systems reported in the literature, including SpeakFaster (SF) [1], Blink-to-Alphabet Tree (B2A) [2], and Webcam Interface (WCI) [5].

TABLE II
FEATURE COMPARISON WITH SURVEYED SYSTEMS

Feature	SF [1]	B2A [2]	WCI [5]	PM
Webcam-only	×	✓	✓	✓
LLM prediction	✓	✓	×	✓
Abbreviation expansion	✓	×	×	✓
Conversation mode	×	×	×	✓
Emergency alert	×	×	×	✓
Caregiver input	×	×	×	✓
Calibration-free	×	✓	×	×
Open-source	✓	N/A	N/A	✓

SF = SpeakFaster, B2A = Blink-to-Alphabet Tree, WCI = Webcam Interface, PM = Proposed Methodology (this work).

As summarized in Table II, the proposed methodology integrates a uniquely complete set of assistive communication capabilities into a single deployable application. While SpeakFaster [1] demonstrated significant motor-action savings through LLM abbreviation expansion, it relied on dedicated infrared hardware. Conversely, webcam-only interfaces such as WCI [5] removed infrared hardware dependencies but lacked predictive language acceleration. The proposed methodology bridges this gap by combining webcam-only tracking with Llama 3.3 70B predictive intelligence, a dual binary-choice conversation engine, rapid-blink SMS alerting, and live caregiver voice/text input. Although B2A [2] provides calibration-free operation, the proposed methodology utilizes a rapid 3-point calibration sequence to achieve higher gaze stability and custom zone adaptation for patients with restricted ocular movement.

B. User Interface Implementation and Workflow

The proposed methodology introduces a modular user interface designed to minimize physical fatigue and support multi-modal communication across various stages of ALS progression.

Fig. 2 depicts the three-point horizontal calibration interface. On application launch, the user focuses on three horizontal screen targets (left, center, right) while 60 gaze samples per target are recorded. Outliers are filtered using interquartile range (IQR) criteria to define patient-specific asymmetric zone thresholds, ensuring robust navigation even under skewed neutral gaze postures.

Fig. 3 displays the primary eye-gaze typing interface. The system employs step-based horizontal navigation (left/right gaze steps with multi-frame hysteresis) rather than continuous cursor control to eliminate tracking jitter. Dwell selection is indicated by a visual countdown progress ring around the highlighted key. The top prediction bar dynamically blends

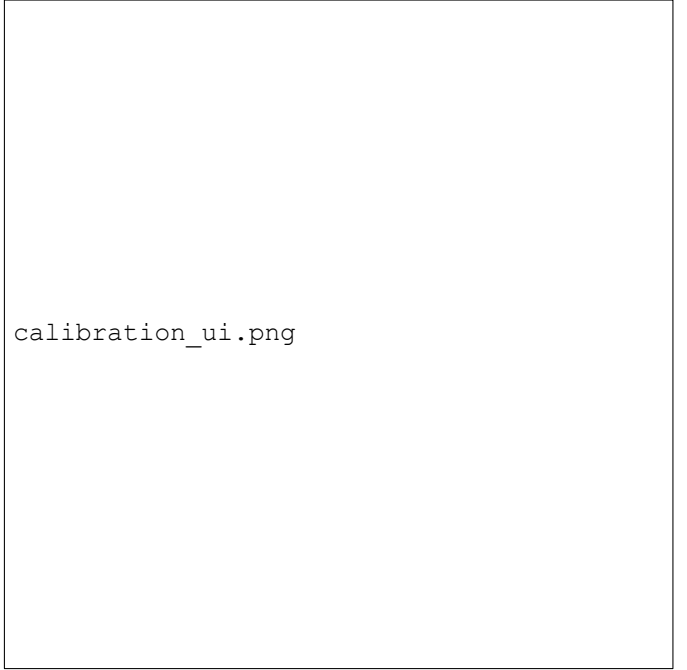


Fig. 2. Three-point horizontal calibration interface with real-time gaze feedback.



Fig. 3. Main eye-gaze QWERTY keyboard interface featuring dwell-time selection ring, prediction slots, and caregiver context bar.

instant local prefix completions with context-aware Llama 3.3 70B suggestions. The bottom context bar allows caregivers to type or speak questions verbally via speech-to-text integration.



Fig. 4. Binary-choice conversation interface presenting dual gaze-selectable response cards generated by Llama 3.3 70B.

Fig. 4 illustrates the binary-choice conversation mode designed for rapid interaction. When a caregiver asks a question (e.g., “do you want water”), the LLM generates two contextual answer options displayed as prominent left/right cards (“Yes please” vs “No thanks”). Looking left or right selects the corresponding option, allowing complex multi-sentence responses to be composed in just 3–5 gaze gestures, significantly reducing motor effort compared to character-by-character typing.

C. Discussion of Design Rationale and System Resilience

The architectural choices of the proposed methodology prioritize accessibility, patient safety, and operational reliability:

- **Offline-First Architecture:** Basic gaze navigation, dwell selection, local prefix matching, offline TTS, and local emergency siren operate fully offline without internet dependency. Cloud connectivity is required only for advanced LLM prediction, abbreviation expansion, binary conversation generation, and Twilio SMS alerts.
- **Safety and Emergency Preparedness:** Moving the rapid-blink emergency alert module directly into system metrics evaluation highlights the reliability of EAR-based blink detection (p_1 – p_6 landmark processing). Triggering a full-screen red visual alert, local siren, and Twilio SMS notification ensures immediate caregiver response during urgent situations.

- **Adaptive Multi-stage Communication:** By supporting letter-by-letter typing, context-aware abbreviation expansion, and binary-choice narrowing within a single interface, the proposed methodology adapts seamlessly as patient fatigue varies throughout the day or as ALS symptoms progress.

VI. CONCLUSION AND FUTURE SCOPE

This paper presented an open-source, webcam-only eye gaze typing system tailored for ALS patients by integrating MediaPipe-based iris tracking, three-point horizontal calibration, step-based gaze navigation, LLM-powered word prediction and abbreviation expansion, a binary-choice conversation engine, rapid-blink emergency SMS alerting, and text-to-speech output. Running entirely on ordinary consumer hardware without dedicated infrared eye-trackers, the proposed methodology addresses key AAC accessibility barriers by offering a multi-modal communication tool that operates across various disease progression stages.

Several limitations remain to be addressed in future revisions. The system has not yet been evaluated in a formal user study with ALS patients, and its throughput has not been formally measured against commercial eye trackers. The three-point calibration required on each launch may present difficulties for patients with very restricted eye movement range, while the step-based navigation model inherently trades maximum typing speed for robustness against tracking noise. Furthermore, the binary conversation engine depends on the LLM generating appropriate and non-harmful options, which is not guaranteed, and the emergency blink alert uses a fixed threshold (five blinks in four seconds) that may need per-patient tuning to avoid false positives in patients with abnormal blink patterns.

Future work will focus on conducting empirical clinical user studies with ALS patients to measure text-entry speed, error rates, and qualitative user satisfaction across different disease stages. We plan to implement adaptive per-patient calibration algorithms for gaze thresholds and blink detection, introduce safety guardrails for LLM-driven response options, and integrate a brain-computer interface (BCI) fallback channel to support patients who transition into a fully locked-in state.

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